

SIGACT News Complexity Theory Column 46

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Introduction to Complexity Theory Column 46

Warmest thanks to Scott Aaronson both for maintaining his wonderful Complexity Zoo (www.complexityzoo.com) and for writing this fascinating article.

Upcoming guest articles include John M. Hitchcock, Jack H. Lutz, and Elvira Mayordomo on *The Fractal Geometry of Complexity Classes*; Neil Immerman on *Recent Progress in Descriptive Complexity*; Subhash Khot on *Recent results in PCPs and Hardness of Approximating NP-Hard Problems*; Piotr Faliszewski and me on *Five Not-So-Easy Questions about the Semifeasible Sets*.

Finally, let me mention that a good way to start the day is by finding that, in your email-box, You've Got Complexity Theorems. And a good way to make that happen is to subscribe to the automatic mailings of new paper abstracts provided by the Computing Research Repository (www.acm.org/repository) and the ECCC (www.eccc.uni-trier.de/eccc). For example, regarding the Computational Complexity part of the Computing Research Repository, to sign up one needs to send an email to "cs@arXiv.org" with (assuming you are Jane Doe) the subject field "subscribe Jane Doe" and with the single line "add Computational Complexity" as the body text. Happy theorems!

Guest Column: NP-complete Problems and Physical Reality

*Scott Aaronson*¹

Abstract

Can NP-complete problems be solved efficiently in the physical universe? I survey proposals including soap bubbles, protein folding, quantum computing, quantum advice, quantum adiabatic algorithms, quantum-mechanical nonlinearities, hidden variables, relativistic time dilation, analog computing, Malament-Hogarth spacetimes, quantum gravity, closed timelike curves, and "anthropic computing." The section on soap bubbles even includes some "experimental" results. While I do not believe that any of the proposals will let us solve NP-complete problems efficiently, I argue that by studying them, we can learn something not only about computation but also about physics.

1 Introduction

"Let a computer smear—with the right kind of quantum randomness—and you create, in effect, a 'parallel' machine with an astronomical number of processors . . . All you have to do is be sure that when you collapse the system, you choose the version that happened to find the needle in the mathematical haystack."

—From *Quarantine* [30], a 1992 science-fiction novel by Greg Egan

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If I had to debate the science writer John Horgan’s claim that basic science is coming to an end [47], my argument would lean heavily on one fact: *it has been only a decade since we learned that quantum computers could factor integers in polynomial time*. In my (unbiased) opinion, the showdown that quantum computing has forced—between our deepest intuitions about computers on the one hand, and our best-confirmed theory of the physical world on the other—constitutes one of the most exciting scientific dramas of our time.

But why did this drama not occur until so recently? Arguably, the main ideas were already in place by the 1960’s or even earlier. I do not know the answer to this sociological puzzle, but can suggest two possibilities. First, many computer scientists see the study of “speculative” models of computation as at best a diversion from more serious work; this might explain why the groundbreaking papers of Simon [66] and Bennett et al. [17] were initially rejected from the major theory conferences. And second, many physicists see computational complexity as about as relevant to the mysteries of Nature as dentistry or tax law.

Today, however, it seems clear that there is something to gain from resisting these attitudes. We would do well to ask: *what else* about physics might we have overlooked in thinking about the limits of efficient computation? The goal of this article is to encourage the serious discussion of this question. For concreteness, I will focus on a single sub-question: *can NP-complete problems be solved in polynomial time using the resources of the physical universe?*

I will argue that studying this question can yield new insights, not just about computer science but about physics as well. More controversially, I will also argue that a negative answer might eventually attain the same status as (say) the Second Law of Thermodynamics, or the impossibility of superluminal signalling. In other words, while experiment will always be the last appeal, the presumed intractability of NP-complete problems might be taken as a useful constraint in the search for new physical theories. Of course, the basic concept will be old hat to computer scientists who live and die by the phrase, “Assuming $P \neq NP$, . . .”

To support my arguments, I will survey a wide range of unusual computing proposals, from soap bubbles and folding proteins to time travel, black holes, and quantum nonlinearities. Some of the proposals are better known than others, but to my knowledge, even the “folklore” ones have never before been collected in one place. In evaluating the proposals, I will try to insist that *all* relevant resources be quantified, and *all* known physics taken into account. As we will see, these straightforward ground rules have been casually ignored in some of the literature on exotic computational models.

Throughout the article, I assume basic familiarity with complexity classes such as P and NP (although not much more than that). Sometimes I do invoke elementary physics concepts, but the difficulty of the physics is limited by my own ignorance.

After reviewing the basics of P versus NP in Section 2, I discuss soap bubbles and related proposals in Section 3, and even report some original “experimental” work in this field. Then Section 4 summarizes what is known about solving NP-complete problems on a garden-variety quantum computer; it includes discussions of black-box lower bounds, quantum advice, and the quantum adiabatic algorithm. Section 5 then considers *variations* on quantum mechanics that might lead to a more powerful model of computation; these include nonlinearities in the Schrödinger equation and certain assumptions about hidden variables. Section 6 moves on to consider analog computing, time dilation, and exotic spacetime geometries; this section is basically a plea to those who think about these matters, to take seriously such trivialities as quantum mechanics and the Planck scale. Relativity and quantum mechanics finally meet in Section 7, on the computational complexity of quantum gravity theories, but the whole point of the section is to explain why this is a premature subject. Sections 8 and 9 finally set aside the more sober ideas (like solving the halting problem using naked singularities), and give zaniness free reign. Section 8 studies the

computational complexity of time travel, while Section 9 studies “anthropic computing,” which means killing yourself whenever a computer fails to produce a certain output. It turns out that even about these topics, there are nontrivial things to be said! Finally, Section 10 makes the case for taking the hardness of NP-complete problems to be a basic fact about the physical world; and weighs three possible objections against doing so.

I regret that, because of both space and cognitive limitations, I was unable to discuss *every* paper related to the solvability of NP-complete problems in the physical world. Two examples of omissions are the gear-based computers of Vergis, Steiglitz, Dickinson [74], and the proposed adiabatic algorithm for the halting problem due to Kieu [53]. Also, I generally ignored papers about “hypercomputation” that did not try to forge *some* link, however tenuous, with the laws of physics as we currently understand them.

2 The Basics

I will not say much about the original P versus NP question: only that the known heuristic algorithms for the 3SAT problem, such as backtrack, simulated annealing, GSAT, and survey propagation, can solve some instances quickly in practice, but are easily stumped by other instances; that the standard opinion is that $P \neq NP$ [40]; that proving this is correctly seen as one of the deepest problems in all of mathematics [50]; that no one has any idea where to begin [34]; and that we have a pretty sophisticated idea of *why* we have no idea [61]. See [68] or [39] for more information.

Of course, even if there is no deterministic algorithm to solve NP-complete problems in polynomial time, there might be a probabilistic algorithm, or a nonuniform algorithm (one that is different for each input length n). But Karp and Lipton [52] showed that either of these would have a consequence, namely the collapse of the polynomial hierarchy, that seems almost as implausible as $P = NP$. Also, Impagliazzo and Wigderson [49] gave strong evidence that $P = BPP$; that is, that any probabilistic algorithm can be simulated by a deterministic one with polynomial slowdown.

It is known that $P \neq NP$ in a “black box” or oracle setting [11]. This just means that any efficient algorithm for an NP-complete problem would have to exploit the problem’s structure in a nontrivial way, as opposed to just trying one candidate solution after another until it finds one that works. Interestingly, most of the physical proposals for solving NP-complete problems that we will see do *not* exploit structure, in the sense that they would still work relative to any oracle. Given this observation, I propose the following challenge: *find a physical assumption under which NP-complete problems can provably be solved in polynomial time, but only in a non-black-box setting.*

3 Soap Bubbles et al.

Given a set of points in the Euclidean plane, a *Steiner tree* (see Figure 1) is a collection of line segments of minimum total length connecting the points, where the segments can meet at vertices (called Steiner vertices) other than the pegs themselves. Garey, Graham, and Johnson [38] showed that finding such a tree is NP-hard.² Yet a well-known piece of computer science folklore maintains that, if two glass plates with pegs between them are dipped into soapy water, then the soap bubbles will rapidly form a Steiner tree connecting the pegs, this being the minimum-energy configuration.

It was only a matter of time before *someone* put the pieces together. Last summer Bringsjord and Taylor [24] posted a paper entitled “P=NP” to the arXiv. This paper argues that, since (1)

²Naturally, the points’ coordinates must be specified to some finite precision. If we only need to decide whether there exists a tree of total length at most L , or whether all trees have length at least $L + \varepsilon$ (for some small $\varepsilon > 0$), then the problem becomes NP-complete.

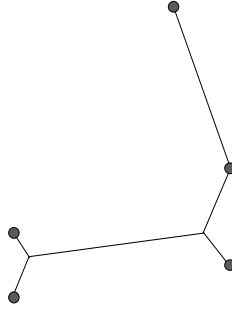


Figure 1: A Steiner tree connecting points at $(.7, .96)$, $(.88, .46)$, $(.88, .16)$, $(.19, .26)$, $(.19, .06)$ (where $(0,0)$ is in the lower left corner, and $(1,1)$ in the upper right). There are two Steiner vertices, at roughly $(.24, .19)$ and $(.80, .26)$.

finding a Steiner tree is NP-hard, (2) soap bubbles find a Steiner tree in polynomial time, (3) soap bubbles are classical objects, and (4) classical physics can be simulated by a Turing machine with polynomial slowdown, it follows that $P = NP$.

My immediate reaction was that the paper was a parody. However, a visit to Bringsjord's home page³ suggested that it was not. Impelled, perhaps, by the same sort of curiosity that causes people to watch reality TV shows, I checked the discussion of this paper on the comp.theory newsgroup to see if anyone recognized the obvious error. And indeed, several posters pointed out that, although soap bubbles might reach a minimum-energy configuration with a small number of pegs, there is no “magical” reason why this should be true in general. By analogy, a rock in a mountain crevice could reach a lower-energy configuration by rolling up first and then down, but it is not observed to do so. A poster named Craig Feinstein replied to these skeptics as follows [33]:

Have experiments been done to show that it is only a local minimum that is reached by soap bubbles and not a global minimum or is this just the party line? I'd like to believe that nature was designed to be smarter than we give it credit. I'd be willing to make a gentleman's bet that no one can site [sic] a paper which describes an experiment that shows that the global minimum is not always achieved with soap bubbles.

Though I was unable to find such a paper, I was motivated by this post to conduct the experiment myself.⁴ I bought two $8'' \times 9''$ glass plates, paint to mark grid points on the plates, thin copper rods which I cut into 1'' pieces, suction cups to attach the rods to the plates, liquid oil soap, a plastic tub to hold the soapy water, and work gloves. I obtained instances of the Euclidean Steiner Tree problem from the OR-Library website [14]. I concentrated on instances with 3 to 7 vertices, for example the one shown in Figure 1.

The result was fascinating to watch: with 3 or 4 pegs, the optimum tree usually *is* found. However, by no means is it always found, especially with more pegs. Soap-bubble partisans might write this off as experimental error, caused (for example) by inaccuracy in placing the pegs, or by the interference of my hands. However, I also sometimes found triangular “bubbles” of three Steiner vertices—which is much harder to explain, since such a structure could never occur in a Steiner tree. In general, the results were highly nondeterministic; I could obtain entirely different

³www.rpi.edu/~brings

⁴S. Aaronson, NP-complete Problems and Physical Reality, *SIGACT News Complexity Theory Column*, March 2005. I win.

trees by dunking the same configuration more than once. Sometimes I even obtained a tree that did not connect all the pegs.

Another unexpected phenomenon was that sometimes the bubbles would start in a suboptimal configuration, then slowly “relax” toward a better one. Even with 4 or 5 pegs, this process could take around ten seconds, and it is natural to predict that with more pegs it would take longer. In short, then, I found no reason to doubt the “party line,” that soap bubbles do not solve NP-complete problems in polynomial time by magic.⁵

There are other proposed methods for solving NP-complete problems that involve relaxation to a minimum-energy state, such as spin glasses and protein folding. All of these methods are subject to the same pitfalls of local optima and potentially long relaxation times. Protein folding is an interesting case, since it seems likely that proteins evolved specifically *not* to have local optima. A protein that folded in unpredictable ways could place whatever organism relied on it at an adaptive disadvantage (although sometimes it happens anyway, as with prions). However, this also means that if we engineered an artificial protein to represent a hard 3SAT instance, then there would be no particular reason for it to fold as quickly or reliably as do naturally occurring proteins.

4 Quantum Computing

Outside of theoretical computer science, parallel computers are sometimes discussed as if they were fundamentally more powerful than serial computers. But of course, anything that can be done with 10^{20} processors in time T can also be done with one processor in time $10^{20}T$. The same is true for DNA strands. Admittedly, for some applications a constant factor of 10^{20} is not irrelevant.⁶ But for solving (say) 3SAT instances with hundreds of thousands of variables, 10^{20} is peanuts.

When quantum computing came along, it was hoped that finally we might have a type of parallelism commensurate with the difficulty of NP-complete problems. For in quantum mechanics, we need a vector of 2^n complex numbers called “amplitudes” just to specify the state of an n -bit computer (see [3, 35, 57] for more details). Surely we could exploit this exponentiality inherent in Nature to try out all 2^n possible solutions to an NP-complete problem in parallel? Indeed, many popular articles on quantum computing have given precisely that impression.

The trouble is that if we measure the computer’s state, we see only *one* candidate solution x , with probability depending on its amplitude α_x . The challenge is to arrange the computation in such a way that only the x ’s we wish to see wind up with large values of α_x . For the special case of factoring, Shor [65] showed that this could be done using a polynomial number of operations—but what about for NP-complete problems?

The short answer is that we don’t know. Indeed, letting BQP be the class of problems solvable in polynomial time by a quantum computer, we do not even know whether $\text{NP} \subseteq \text{BQP}$ would imply $\text{P} = \text{NP}$ or some other unlikely consequence in classical complexity.⁷ But in 1994, Bennett, Bernstein, Brassard, and Vazirani [17] did show that $\text{NP} \not\subseteq \text{BQP}$ relative to an oracle. In particular, they showed that any quantum algorithm that searches an unordered database of N items for a single “marked” item must query the database $\sim \sqrt{N}$ times. (Soon afterward, Grover [43] showed that this is tight.)

⁵Some people have objected that, while all of this might be true in practice, I still have not shown that soap bubbles cannot solve NP-complete problems *in principle*. But what exactly does “in principle” mean? If it means obeying the equations of classical physics, then the case for magical avoidance of local optima moves from empirically weak to demonstrably false, as in the case of the rock stuck in the mountain crevice.

⁶This is one fact I seem to remember from my computer architecture course.

⁷On the other hand, if #P-complete problems were solvable in quantum polynomial time, then this *would* have an unlikely classical complexity consequence, namely the collapse of the so-called counting hierarchy.

If we interpret the space of 2^n possible assignments to a Boolean formula φ as a “database,” and the satisfying assignments of φ as “marked items,” then Bennett et al.’s result says that any quantum algorithm needs at least $\sim 2^{n/2}$ steps to find a satisfying assignment of φ with high probability, *unless* the algorithm exploits the structure of φ in a nontrivial way. In other words, there is no “brute-force” quantum algorithm to solve NP-complete problems in polynomial time, just as there is no brute-force classical algorithm.

In Bennett et al.’s original proof, we first run our quantum algorithm on a database with no marked items. We then mark the item that was queried with the smallest total probability, and show that the algorithm will need many queries to notice this change. By now, many other proofs have been discovered, including that of Beals et al. [13], which represents an efficient quantum algorithm’s acceptance probability by a low-degree polynomial, and then shows that no such polynomial exists; and that of Ambainis [9], which upper-bounds how much the entanglement between the algorithm and database can increase via a single query. Both techniques have also led to lower bounds for many other problems besides database search.

The crucial property of quantum mechanics that all three proofs exploit is its *linearity*: the fact that, until a measurement is made, the vector of amplitudes can only evolve by means of linear transformations. Intuitively, if we think of the components of a superposition as “parallel universes,” then linearity is what prevents the universe containing the marked item from simply “telling all the other universes about it.”

4.1 Quantum Advice

The above assumed that our quantum computer begins in some standard initial state, such as the “all-0” state (denoted $|0 \cdots 0\rangle$). An interesting twist is to consider the effects of other initial states. Are there quantum states that could take exponential time to prepare, but that would let us solve NP-complete problems in polynomial time were they given to us by a wizard? More formally, let BQP/qpoly be the class of problems solvable in quantum polynomial time, given a polynomial-size “quantum advice state” $|\psi_n\rangle$ that depends only on the input length n . Then recently I showed that $\text{NP} \not\subseteq \text{BQP/qpoly}$ relative to an oracle [2]. Intuitively, even if the state $|\psi_n\rangle$ encoded the solutions to every 3SAT instance of size n , only a miniscule fraction of that information could be extracted by measuring $|\psi_n\rangle$, at least within the black-box model that we know how to analyze. The proof uses the polynomial technique of Beals et al. [13] to prove a so-called *direct product theorem*, which upper-bounds the probability of solving many database search problems simultaneously. It then shows that this direct product theorem could be violated, if the search problem were efficiently solvable using quantum advice.

4.2 The Quantum Adiabatic Algorithm

At this point, some readers may be getting impatient with the black-box model. After all, NP-complete problems are *not* black boxes, and classical algorithms such as backtrack search do exploit their structure. Why couldn’t a quantum algorithm do the same? A few years ago, Farhi et al. [32] announced a new *quantum adiabatic algorithm*, which can be seen as a quantum analogue of simulated annealing. Their algorithm is easiest to describe in a continuous-time setting, using the concepts of a Hamiltonian (an operation that acts on a quantum state over an infinitesimal time interval Δt) and a ground state (the lowest-energy state left invariant by a given Hamiltonian). The algorithm starts by applying a Hamiltonian H_0 that has a known, easily prepared ground state, then slowly transitions to another Hamiltonian H_1 whose ground state encodes the solution to (say) an instance of 3SAT. The *quantum adiabatic theorem* says that if a quantum computer

starts in the ground state of H_0 , then it must end in the ground state of H_1 , *provided the transition from H_0 to H_1 is slow enough*. The key question is how slow is slow enough.

In their original paper, Farhi et al. [32] gave numerical evidence that the adiabatic algorithm solves random, critically-constrained instances of the NP-complete Exact Cover problem in polynomial time. But having learned from experience, most computer scientists are wary of taking such numerical evidence too seriously as a guide to asymptotic behavior. This is especially true when the instance sizes are small ($n \leq 20$ in Farhi et al.'s case), as they have to be when simulating a quantum computer on a classical one. On the other hand, Farhi relishes pointing out that if the empirically-measured running time were exponential, no computer scientist would dream of saying that it would eventually become polynomial! In my opinion, the crucial experiment (which has not yet been done) would be to compare the adiabatic algorithm head-on against simulated annealing and other classical heuristics. The evidence for the adiabatic algorithm's performance would be much more convincing if the known classical algorithms took exponential time on the same random instances.

On the theoretical side, van Dam, Mosca, and Vazirani [73] constructed 3SAT instances for which the adiabatic algorithm provably takes exponential time, at least when the transition between the initial and final Hamiltonians is linear. Their instances involve a huge "basin of attraction" that leads to a false optimum (meaning most but not all clauses are satisfied), together with an exponentially small basin that leads to the true optimum. To lower-bound the algorithm's running time on these instances, van Dam et al. showed that the spectral gap (that is, the gap between the smallest and second-smallest eigenvalues) of some intermediate Hamiltonian decreases exponentially in n . As it happens, physicists have almost a century of experience in analyzing these spectral gaps, but not for the purpose of deciding whether they decrease polynomially or exponentially as the number of particles increases to infinity!

Such "hands-on" analysis of the adiabatic algorithm was necessary, since van Dam et al. also showed that there is no black-box proof that the algorithm takes exponential time. This is because, given a variable assignment X to the 3SAT instance φ , the adiabatic algorithm computes not merely *whether* X satisfies φ , but also how many clauses it satisfies. And this information turns out to be sufficient to reconstruct φ itself.

Recently Reichardt [62], building on work of Farhi, Goldstone, and Gutmann [31], has constructed 3SAT instances for which the adiabatic algorithm takes polynomial time, whereas simulated annealing takes exponential time. These instances involve a narrow obstacle along the path to the global optimum, which simulated annealing gets stuck at but which the adiabatic algorithm tunnels past. On the other hand, these instances are easily solved by other classical algorithms. An interesting open question is whether there exists a family of black-box functions $f : \{0, 1\}^n \rightarrow \mathbb{Z}$ for which the adiabatic algorithm finds a global minimum using exponentially fewer queries than *any* classical algorithm.

5 Variations on Quantum Mechanics

Quantum computing skeptics sometimes argue that we do not *really* know whether quantum mechanics itself will remain valid in the regime tested by quantum computing.⁸ Here, for example, is Leonid Levin [55]: "The major problem [with quantum computing] is the requirement that basic quantum equations hold to multi-hundredth if not millionth decimal positions where the significant digits of the relevant quantum amplitudes reside. We have never seen a physical law valid to over a dozen decimals."

⁸Personally, I agree, and consider this the main motivation for trying to build a quantum computer.

The irony is that most of the specific proposals for how quantum mechanics *could* be wrong suggest a world with more, not less, computational power than BQP. For, as we saw in Section 4, the linearity of quantum mechanics is what prevents one needle in an exponentially large haystack from shouting above the others. And as observed by Weinberg [76], it seems difficult to change quantum mechanics in any consistent way while preserving linearity.

But how drastic could the consequences possibly be, if we added a tiny nonlinear term to the Schrödinger equation (which describes how quantum states evolve in time)? For starters, Gisin [41] and Polchinski [59] showed that in most nonlinear variants of quantum mechanics, one could use entangled states to transmit superluminal signals. More relevant for us, Abrams and Lloyd [6] showed that one could solve NP-complete and even #P-complete problems in polynomial time—at least if the computation were error-free. Let us see why this is, starting with NP.

Given a black-box function f that maps $\{0, 1\}^n$ to $\{0, 1\}$, we want to decide in polynomial time whether there exists an input x such that $f(x) = 1$. We can start by preparing a uniform superposition over all inputs, denoted $2^{-n/2} \sum_x |x\rangle$, and then querying the oracle for f , to produce $2^{-n/2} \sum_x |x\rangle |f(x)\rangle$. If we then apply Hadamard gates to the first register and measure that register, one can show that we will obtain the outcome $|0 \cdots 0\rangle$ with probability at least $1/4$. Furthermore, conditioned on the first register having the state $|0 \cdots 0\rangle$, the second register will be in the state

$$\frac{(2^n - s) |0\rangle + s |1\rangle}{\sqrt{(2^n - s)^2 + s^2}}$$

where s is the number of inputs x such that $f(x) = 1$. So the problem reduces to that of distinguishing two possible states of a single qubit—for example, the states corresponding to $s = 0$ and $s = 1$. The only difficulty is that these states are exponentially close.

But a nonlinear operation need not preserve the angle between quantum states—it can pry them apart. Indeed, Abrams and Lloyd showed that by repeatedly applying a particular kind of nonlinear gate, which arises in a model of Weinberg [76], one could increase the angle between two quantum states *exponentially*, and thereby distinguish the $s = 0$ and $s = 1$ cases with constant bias. It seems likely that “almost any” nonlinear gate would confer the same ability, though it is unclear how to formalize this statement.

To solve #P-complete problems, we use the same basic algorithm, but apply it repeatedly to “zoom in” on the value of s using binary search. Given any range $[a, b]$ that we believe contains s , by applying the nonlinear gate a suitable number of times we can make the case $s = a$ correspond roughly to $|0\rangle$, and the case $s = b$ correspond roughly to $|1\rangle$. Then measuring the state will provide information about whether s is closer to a or b . This is true even if $(b - a)/2^n$ is exponentially small.

Indeed, if arbitrary 1-qubit nonlinear operations are allowed, then it is not hard to see that we could even solve PSPACE-complete problems in polynomial time. It suffices to solve the following problem: given a Boolean function f of n bits $x_1, \dots, x_n$, does there exist a setting of x_1 such that for all settings of x_2 there exists a setting of x_3 such that... $f(x_1, \dots, x_n) = 1$? To solve this, we can first prepare the state

$$\frac{1}{2^{n/2}} \sum_{x_1, \dots, x_n} |x_1 \dots x_n, f(x_1 \dots x_n)\rangle.$$

We then apply a “nonlinear AND gate” to the n^{th} and $(n + 1)^{\text{st}}$ qubits, which maps $|00\rangle + |10\rangle$, $|00\rangle + |11\rangle$, and $|01\rangle + |10\rangle$ to $|00\rangle + |10\rangle$, and $|01\rangle + |11\rangle$ to itself (omitting the $\sqrt{2}$ normalization). Next we apply a “nonlinear OR gate” to the $(n - 1)^{\text{st}}$ and $(n + 1)^{\text{st}}$ qubits, which maps $|00\rangle + |11\rangle$, $|01\rangle + |10\rangle$, and $|01\rangle + |11\rangle$ to $|01\rangle + |11\rangle$, and $|00\rangle + |10\rangle$ to itself. We continue to alternate between

AND and OR in this manner, while moving the control qubit leftward towards x_1 . At the end, the $(n+1)^{st}$ qubit will be $|1\rangle$ if the answer is ‘yes,’ and $|0\rangle$ if the answer is ‘no.’

On the other hand, any nonlinear quantum computer can also be simulated in PSPACE. For even in nonlinear theories, the amplitude of any basis state at time t is an easily-computable function of a small number of amplitudes at time $t-1$, and can therefore be computed in polynomial space using depth-first recursion. It follows that, assuming arbitrary nonlinear gates and no error, PSPACE exactly characterizes the power of nonlinear quantum mechanics.

But what if we allow error, as any physically reasonable model of computation must? In this case, while it *might* still be possible to solve NP-complete problems in polynomial time, I am not convinced that Abrams and Lloyd have demonstrated this.⁹ Observe that the standard quantum error-correction theorems break down, since just as a tiny probability of success can be magnified exponentially during the course of a computation, so too can a tiny probability of error. Whether this problem can be overcome might depend on which specific nonlinear gates are available; the issue deserves further investigation.

5.1 Hidden-Variable Theories

Most people who quote Einstein’s declaration that “God does not play dice” seem not to realize that a dice-playing God would be an *improvement* over the actual situation. In quantum mechanics, a particle does not have a position, even an unknown position, until it is measured. This means that it makes no sense to talk about a “trajectory” of the particle, or even a probability distribution over possible trajectories. And without such a distribution, it is not clear how we can make even probabilistic predictions for future observations, if we ourselves belong to just one component of a larger superposition.

Hidden-variable theories try to remedy this problem by supplementing quantum mechanics with the “actual” values of certain observables (such as particle positions or momenta), together with rules for how those observables evolve in time. The most famous such theory is due to Bohm [20], but there are many alternatives that are equally compatible with experiment. Indeed, a key feature of hidden-variable theories is that they reproduce the usual quantum-mechanical probabilities at any individual time, and so are empirically indistinguishable from ordinary quantum mechanics. It does not seem, therefore, that a “hidden-variable quantum computer” could possibly be more powerful than a garden-variety one.

On the other hand, it might be that Nature needs to “expend more computational effort” to calculate a particle’s entire trajectory than to calculate its position at any individual time. The reason is that the former requires keeping track of multiple-time correlations. And indeed, I showed in [4] that under any hidden-variable theory satisfying a reasonable axiom called “indifference to the identity,” the ability to sample the hidden variable’s history would let us solve the Graph Isomorphism problem in polynomial time. For intuitively, given two graphs G and H with no nontrivial automorphisms, one can easily prepare a uniform superposition over all permutations of G and H :

$$\frac{1}{\sqrt{2n!}} \sum_{\sigma \in S_n} (|0\rangle |\sigma\rangle |\sigma(G)\rangle + |1\rangle |\sigma\rangle |\sigma(H)\rangle).$$

Then measuring the third register yields a state of the form $|i\rangle |\sigma\rangle$ if G and H are not isomorphic, or $(|0\rangle |\sigma\rangle + |1\rangle |\tau\rangle) / \sqrt{2}$ for some $\sigma \neq \tau$ if they are isomorphic. Unfortunately, if then we measured

⁹Abrams and Lloyd claimed to give an algorithm that does not require “exponentially precise operations.” The problem is that their algorithm uses a nonlinear OR gate, and depending on how that gate behaves on states other than $|00\rangle + |10\rangle$, $|00\rangle + |11\rangle$, $|01\rangle + |10\rangle$, and $|01\rangle + |11\rangle$, it might magnify small errors exponentially. In particular, I could not see how to implement a nonlinear OR gate robustly using Abrams and Lloyd’s “Weinberg gate.”

this state in the standard basis, we would get no information whatsoever, and work of myself [1], Shi [64], and Midrijanis [56] shows that no black-box quantum algorithm can do much better. But if only we could make a few “non-collapsing” measurements! Then we would see the same permutation each time in the former case, but two permutations with high probability in the latter.

The key point is that seeing a hidden variable’s history would effectively let us simulate non-collapsing measurements. Using this fact, I showed that by sampling histories, we could simulate the entire class SZK of problems having statistical zero-knowledge proofs, which includes Graph Isomorphism, Approximate Shortest Vector, and other NP-intermediate problems for which no efficient quantum algorithm is known. On the other hand, SZK is not thought to contain the NP-complete problems; indeed, if it did then the polynomial hierarchy would collapse [21]. And it turns out that, even if we posit the unphysical ability to sample histories, we *still* could not solve NP-complete problems efficiently in the black-box setting! The best we could do is search a list of N items in $\sim N^{1/3}$ steps, as opposed to $\sim N^{1/2}$ with Grover’s algorithm.

But even if a hidden-variable picture is correct, are these considerations relevant to any computations *we* could perform? They would be, if a proposal of Valentini [72, 71] were to pan out. Valentini argues that the $|\psi|^2$ probability law merely reflects a statistical equilibrium (analogous to thermal equilibrium), and that it might be possible to find “nonequilibrium matter” (presumably left over from the Big Bang) in which the hidden variables still obey a different distribution. Using such matter, Valentini showed that we could distinguish nonorthogonal states, and thereby transmit superluminal signals and break quantum cryptographic protocols. He also claimed that we could solve NP-complete problems in polynomial time. Unfortunately, his algorithm involves measuring a particle’s position to exponential precision, and if we could do that, then it is unclear why we could not also solve NP-complete problems in polynomial time *classically*! So in my view, the power of Valentini’s model with realistic constraints on precision remains an intriguing open question. My conjecture is that it will turn out to be similar to the power of the histories model—that is, able to solve SZK problems in polynomial time, but not NP-complete problems in the black-box setting. I would love to be disproven.

6 Relativity and Analog Computing

If quantum computers cannot solve NP-complete problems efficiently, then perhaps we should turn to the other great theory of twentieth-century physics: relativity. The idea of relativity computing is simple: start your computer working on an intractable problem, then board a spaceship and accelerate to nearly the speed of light. When you return to Earth, all of your friends will be long dead, but the answer to your problem will await you.

What is the problem with this proposal? Ignoring the time spent accelerating and decelerating, if you travelled at speed v relative to Earth for proper time t (where $v = 1$ is light speed), then the elapsed time in your computer’s reference frame would be $t' = t/\sqrt{1 - v^2}$. It follows that, if you want t' to increase exponentially with t , then v has to be exponentially close to the speed of light. But this implies that the amount of *energy* needed to accelerate the spaceship also increases exponentially with t . So your spaceship’s fuel tank (or whatever else is powering it) will need to be exponentially large—which means that you will again need exponential time, just for the fuel from the far parts of the tank to affect you!

Similar remarks apply to traveling close to a black hole event horizon: if you got exponentially close then you would need exponential energy to escape.¹⁰ On the other hand, Malament and

¹⁰An interesting property of relativity is that it is always *you* who has to go somewhere or do something in these proposals, while the computer stays behind. Conversely, if you wanted more time to think about what to say next

Hogarth (see [46]) have constructed spacetimes in which, by traveling for a finite proper time along one worldline, an observer could see the *entire infinite past* of another worldline. Naturally, this would allow that observer to solve not only NP-complete problems but the halting problem as well. It is known that these spacetimes cannot be globally hyperbolic; that is to say, they must have naked singularities, which are points at which general relativity no longer yields predictions. But to me, the mere existence of such singularities is a relatively minor problem, since there is evidence today that they really *can* form in classical general relativity (see [67] for a survey).

The real problem is the Planck scale. By combining three physical constants—Planck’s constant $\hbar \approx 1.05 \times 10^{-34} m^2 kg^1 s^{-1}$, Newton’s gravitational constant $G \approx 6.67 \times 10^{-11} m^3 kg^{-1} s^{-2}$, and the speed of light $c \approx 3.00 \times 10^8 m^1 kg^0 s^{-1}$ —one can obtain a fundamental unit of length known as the *Planck length*:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.62 \times 10^{-35} m.$$

The physical interpretation of this length is that, if we tried to confine an object inside a sphere of diameter ℓ_P , then the object would acquire so much energy that it would collapse to form a black hole. For this reason, most physicists consider it meaningless to discuss lengths shorter than the Planck length, or times shorter than the corresponding Planck time $\ell_P/c \approx 5.39 \times 10^{-44} s$. They assume that, even if there do exist naked singularities, these are simply places where general relativity breaks down on length scales of order ℓ_P , and must be replaced by a quantum theory of gravity.

Indeed, Bekenstein [16] gave an upper bound on the total information content of any isolated, weakly gravitating physical system, by assuming the Second Law of Thermodynamics and then considering a thought experiment in which the system is slowly lowered into a black hole. Specifically, he showed that $S \leq 2\pi ER$, where S is the entropy of the system, or $\ln 2$ times the number of bits of information; E is the system’s gravitating energy; and R is the radius of the smallest sphere containing the system. Note that E and R are in Planck units. Since the energy of a system can be at most proportional to its radius (at least according to the widely-believed “hoop conjecture”), one corollary of Bekenstein’s bound is the *holographic bound*: the information content of any region is at most proportional to the surface area of the region, at a rate of one bit per Planck length squared, or 1.4×10^{69} bits per square meter. Bousso [23], whose survey paper on this subject is well worth reading by computer scientists, has reformulated the holographic bound in a generally covariant way, and marshaled a surprising amount of evidence for its validity.

Some physicists go even further, and maintain that space and time are literally discrete on the Planck scale. Of course, the discreteness could not be of the straightforward kind that occurs in cellular automata such as Conway’s Game of Life, since that would fail to reproduce Lorentz or even Galilean invariance. Instead, it would be a more subtle, quantum-mechanical kind of discreteness, as appears for example in loop quantum gravity (see Section 7). But I should stress that the holographic bound itself, and the existence of a Planck scale at which classical ideas about space and time break down, are generic conclusions that stand independently of any specific quantum gravity theory.

The reason I have taken this detour into Planck-scale physics is that our current understanding seems to rule out, not only the Malament-Hogarth proposal, but *all similar proposals* for solving the halting problem in finite time. Yet in the literature on “hypercomputation” [28, 46], one still reads about machines that could “bypass the Turing barrier” by performing the first step of a computation in one second, the second in 1/2 second, the third in 1/4 second, and so on, so that after two seconds an infinite number of steps has been performed. Sometimes the proposed

in a conversation, then your conversational partner is the one who would have to be placed in a spaceship.

mechanism invokes Newtonian physics (ignoring even the finiteness of the speed of light), while other times it requires traveling arbitrarily close to a spacetime singularity. Surprisingly, in the papers that I encountered, the most common response to quantum effects was not to discuss them at all!

The closest I found to an account of physicality comes from Hogarth [46], who stages an interesting dialogue between a traditional computability theorist named Frank and a hypercomputing enthusiast named Isabel. After Isabel describes a type of spacetime that would support “non-Turing computers,” the following argument ensues:

Frank: Yes, but surely the spacetime underlying our universe is not like that. These solutions [to Einstein’s equation] are just idealisations.

Isabel: That’s beside the point. You don’t want to rubbish a hypothetical computer—Turing or non-Turing—simply because it can’t fit into our universe. If you do, you’ll leave your precious Turing machine to the mercy of the cosmologists, because according to one of their theories, the universe and all it contains, will crunch to nothing in a few billion years. Your Turing machine would be cut-off in mid-calculation! [46, p. 15]

I believe that Isabel’s analogy fails. For in principle, one can generally translate theorems about Turing machines into statements about what Turing computers could or could not do within the time and space bounds of the physical universe.¹¹ By contrast, it is unclear if claims about hypercomputers have *any* relevance whatsoever to the physical universe. The reason is that, if the n^{th} step of a hypercomputation took 2^{-n} seconds, then it would take fewer than 150 steps to reach the Planck time.

In my view, the “foaminess” of space and time on the Planck scale also rules out approaches to NP-complete problems based on analog computing. (For present purposes, an analog computer is a machine that performs a discrete sequence of steps, but on unlimited-precision real numbers.) As an example of such an approach, in 1979 Schönhage [63] showed how to solve NP-complete and even PSPACE-complete problems in polynomial time, given the ability to compute $x + y$, $x - y$, xy , x/y , and $\lfloor x \rfloor$ in a single time step for any two real numbers x and $y \neq 0$. Intuitively, one can use the first $2^{\Theta(n)}$ bits in a real number’s binary expansion to encode an instance of the Quantified Boolean Formula problem, then use arithmetic operations to calculate the answer in parallel, and finally extract the binary result.¹² The problem, of course, is that unlimited-precision real numbers would violate the holographic entropy bound.

7 Quantum Gravity

Here we enter a realm of dragons, where speculation abounds but concrete ideas about computation are elusive. The one clear result is due to Freedman, Kitaev, Larsen, and Wang [36, 37], who studied topological quantum field theories (TQFT’s). These theories, which arose from the work of Witten and others in the 1980’s, involve 2 spatial dimensions and 1 time dimension. Dropping from 3 to 2 dimensions might seem like a trivial change to a computer scientist, but it has the effect of making quantum gravity radically simpler; basically, the only degree of freedom is now the topology of

¹¹As an example, Stockmeyer and Meyer [70] gave a simple problem in logic, such that solving instances of size 610 provably requires circuits with at least 10^{125} gates.

¹²Note that the ability to apply the floor function (or equivalently, to access a specific bit in a real number’s binary expansion) is essential here. If we drop that ability, then we obtain the beautiful theory of algebraic complexity [18, 26], which has its own “P versus NP” questions over the real and complex numbers. These questions are logically unrelated to the original P versus NP question so far as anyone knows—possibly they are easier.

the spacetime manifold, together with any “punctures” in that manifold. Surprisingly, Freedman et al. were able to define a model of computation based on TQFT’s, and show that this model is equivalent to ordinary quantum computation: more precisely, all TQFT’s can be simulated in BQP, and some TQFT’s are universal for BQP. Unfortunately, the original papers on this discovery are all but impossible for a computer scientist to read, but Aharonov, Jones, and Landau [7] are currently working on a simplified presentation.

From what I understand, it remains open to analyze the computational complexity of $(3 + 1)$ -dimensional quantum field theories even in flat spacetime. Part of the problem is that these theories are not mathematically rigorous: they have well-known infinities, which are swept under the rug via a process called “renormalization.” However, since the theories in some sense preserve quantum-mechanical unitarity, the expectation of physicists I have asked is that they will not lead to a model of computation more powerful than BQP.

The situation is different for speculative theories incorporating gravity, such as M-theory, the latest version of string theory. For these theories involve a notion of “locality” that is much more subtle than the usual one: in particular, the so-called AdS/CFT correspondence proposes that theories with gravity in d dimensions are somehow isomorphic to theories without gravity in $d - 1$ dimensions (see [19]). As a result, Preskill [60] has pointed out that even if M-theory remains based on standard quantum mechanics, it might allow the efficient implementation of unitary transformations that would require exponential time on an ordinary quantum computer. It would be interesting to develop this idea further.

String theory’s main competitor is a theory called loop quantum gravity.¹³ Compared to string theory, loop quantum gravity has one feature that I find attractive as a computer scientist: it explicitly models spacetime as discrete and combinatorial on the Planck scale. In particular, one can represent the states in this theory by sums over *spin networks*, which are undirected graphs with edges labeled by integers. The spin networks evolve via local operations called Pachner moves; a sequence of these moves is called a spin foam. Then the “amplitude” for transitioning from spin network A to spin network B equals the sum, over all spin foams F going from A to B , of the amplitude of F . In a specific model known as the Riemannian¹⁴ Barrett-Crane model, this amplitude equals the product, over all Pachner moves in F , of an expression called a “10j symbol,” which can be evaluated according to rules originally developed by Penrose [58].

Complicated, perhaps, but this seems like the stuff out of which a computational model could be made. So two years ago I spoke with Dan Christensen, a mathematician who along with Greg Egan gave an efficient algorithm [27] for calculating 10j symbols that has been crucial in the numerical study of spin foams. I wanted to know whether one could define a complexity class “BQGP” (Bounded-Error Quantum Gravity Polynomial-Time) based on spin foams, and if so, how it compared to BQP. The first observation we made is that evaluating arbitrary spin networks (as opposed to 10j symbols) using Penrose’s rules is $\#P$ -complete. This follows by a simple reduction from counting the number of edge 3-colorings of a trivalent planar graph, which was proven $\#P$ -complete by Vertigan and Welsh [75].

But what about simulating the dynamics of (say) the Barrett-Crane model? Here we quickly ran into problems: for example, in summing over all spin foams between two spin networks, should one impose an upper bound on the number of Pachner moves, and if so, what? Also, supposing we *could* compute amplitudes for transitioning from one spin network to another, what would these

¹³If some physicist wants to continue the tradition of naming quantum gravity theories using monosyllabic words for elongated objects that mean something completely different in computer science, then I propose the most revolutionary advance yet: *thread theory*.

¹⁴Here “Riemannian” means not taking into account that time is different from space. There is also a Lorentzian Barrett-Crane model, but it is considerably more involved.

numbers represent? If they are supposed to be analogous to transition amplitudes in ordinary quantum mechanics, then how do we normalize them so that probabilities sum to unity? In the quantum gravity literature, issues such as these are still not settled.¹⁵

In the early days of quantum mechanics, there was much confusion about the operational meaning of the wavefunction. (Even in Born’s celebrated 1926 paper [22], the idea that one has to *square* amplitudes to get probabilities only appeared in a footnote added in press!) Similarly, Einstein struggled for years to extract testable physics from a theory in which any coordinate system is as valid as any other. So maybe it is no surprise that, while today’s quantum gravity researchers can write down equations, they are still debating what seem to an outsider like extremely basic questions about what the equations mean. The trouble is that these questions are exactly the ones we need answered, if we want to formulate a model of computation! Indeed, to anyone who wants a test or benchmark for a favorite quantum gravity theory,¹⁶ let me humbly propose the following: *can you define Quantum Gravity Polynomial-Time?*

A possible first step would be to define time. For in many quantum gravity theories, there is not even a notion of objects evolving dynamically in time: instead there is just a static spacetime manifold, subject to a constraint such as the Wheeler-DeWitt equation $H\Psi = 0$. In classical general relativity, at least we could carve the universe into ‘spacelike slices’ if we wanted to, and assign a local time to any given observer! But how do we do either of those if the spacetime metric itself is in quantum superposition? Regulars call this “the problem of time” (see [69] for a fascinating discussion). The point I wish to make is that, until this and the other conceptual problems have been clarified—until we can say what it means for a ‘user’ to specify an ‘input’ and ‘later’ receive an ‘output’—*there is no such thing as computation, not even theoretically.*

8 Time Travel Computing

Having just asserted that a concept of time something like the usual one is needed even to define computation, I am now going to disregard that principle, and discuss computational models that exploit closed timelike curves (CTC’s). The idea was well explained by the movie *Star Trek IV: The Voyage Home*. The Enterprise crew has traveled back in time to the present (meaning to 1986) in order to find humpback whales and bring them into the twenty-third century. The problem is that building a tank to transport the whales requires a type of plexiglass that has not yet been invented. In desperation, the crew seeks out the company that *will* invent the plexiglass, and reveals its molecular formula to that company. The question is, where did the work of inventing the formula take place?

In a classic paper on CTC’s, Deutsch [29] observes that, in contrast to the much better-known grandfather paradox, the “knowledge creation paradox” involves no logical contradiction. The only paradox is a complexity-theoretic one: a difficult computation somehow gets performed, yet without the expected resources being devoted to it. Deutsch goes further, and argues that this is *the* paradox of time travel, the other ones vanishing once quantum mechanics is taken into account. The idea is this: consider a unitary matrix U acting on the Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, where $\mathcal{H}_A$ consists of ‘chronology-respecting qubits’ and $\mathcal{H}_B$ consists of ‘closed timelike curve qubits’ (see Figure 2). Then one can show that there always exists a mixed quantum state ρ of the $\mathcal{H}_B$ qubits, such that if we start with $|0 \cdots 0\rangle$ in $\mathcal{H}_A$ and ρ in the $\mathcal{H}_B$, apply U , and then trace out $\mathcal{H}_A$, the resulting state in $\mathcal{H}_B$ is again ρ . Deutsch calls this requirement *causal consistency*. What it means

¹⁵If the normalization were done manually, then presumably one could solve NP-complete problems in polynomial time using postselection (see Section 9). This seems implausible.

¹⁶That is, one without all the bother of making numerical predictions and comparing them to observation.

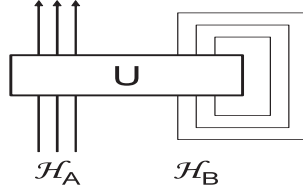


Figure 2: Deutsch’s causal consistency model consists of ‘chronology-respecting qubits’ in Hilbert space $\mathcal{H}_A$, and ‘CTC qubits’ in Hilbert space $\mathcal{H}_B$ whose quantum state must be invariant under U .

is that ρ is a *fixed point* of the superoperator¹⁷ acting on $\mathcal{H}_B$, so we can take it to be both the ‘input’ and ‘output’ of the CTC.

Strictly speaking, Deutsch’s idea does not depend on quantum mechanics; we could equally well say that any Markov chain has a stationary distribution. In both the classical and quantum cases, the resolution of the grandfather paradox is then that you are born with 1/2 probability, and *if* you are born you go back in time to kill your grandfather, from which it follows that you are born with 1/2 probability, and so on.

One advantage of this resolution is that it immediately suggests a model of computation. For simplicity, let us first consider the classical case, and assume *all* bits go around the CTC (this assumption will turn out not to matter for complexity purposes). Then the model is the following: first the user specifies as input a polynomial-size circuit $C : \{0, 1\}^n \rightarrow \{0, 1\}^n$. Then Nature chooses a probability distribution $\mathcal{D}$ over $\{0, 1\}^n$ that is left invariant by C . Finally, the user receives as output a sample x from $\mathcal{D}$, which can be used as the basis for further computation. An obvious question is, if there is more than one stationary distribution $\mathcal{D}$, then which one does Nature choose? The answer turns out to be irrelevant, since we can construct circuits C such that a sample from *any* stationary distribution could be used to solve NP-complete or even PSPACE-complete problems in polynomial time.

The circuit for NP-complete problems is simple: given a Boolean formula φ , let $C(x) = x$ if x is a satisfying assignment for φ , and $C(x) = x + 1$ otherwise, where x is considered as an n -bit integer and the addition is mod 2^n . Then provided φ has any satisfying assignments at all, the only stationary distributions of C will be the singleton distributions concentrated on those assignments.

I am indebted to Lance Fortnow for coming up with a time travel circuit for the more general case of PSPACE-complete problems. Let $M_1, \dots, M_T$ be the successive configurations of a PSPACE machine M . Then our circuit C will take as input a machine configuration M_t together with a bit $i \in \{0, 1\}$. The circuit does the following: if $t < T$, then C maps each (M_t, i) to (M_{t+1}, i) . Otherwise, if $t = T$, then C maps (M_T, i) to $(M_1, 0)$ if M_T is a rejecting state, or (M_T, i) to $(M_1, 1)$ if M_T is an accepting state. Notice that if M accepts, then the only stationary distribution of C is the uniform distribution over the cycle $\{(M_1, 1), \dots, (M_T, 1)\}$. On the other hand, if M rejects, then the only stationary distribution is uniform over $\{(M_1, 0), \dots, (M_T, 0)\}$. So in either case, measuring i yields the desired output.

Conversely, it is easy to see that a PSPACE machine can sample from some stationary distri-

¹⁷A “superoperator” is a generalization of a unitary matrix that can include interaction with ancilla qubits, and therefore need not be reversible.

bution of C . For the problem reduces to finding a cycle in the exponentially large graph of the function $C : \{0, 1\}^n \rightarrow \{0, 1\}^n$, and then choosing a uniform random vertex from that cycle. The same idea works even if not all n of the bits go around the CTC. It follows that PSPACE exactly characterizes the classical computational complexity of time travel, if we assume Deutsch's causal consistency requirement.

But what about the *quantum* complexity of time travel? The model is as follows: first the user specifies a polynomial-size quantum circuit C acting on $\mathcal{H}_A \otimes \mathcal{H}_B$; then Nature adversarially chooses a mixed state ρ such that $\text{Tr}_A [C(|0 \cdots 0\rangle \langle 0 \cdots 0| \otimes \rho)] = \rho$, where Tr_A denotes partial trace over $\mathcal{H}_A$; and finally the user can perform an arbitrary BQP computation on ρ . Let BQP_{CTC} be the class of problems solvable in this model. Then it is easy to see that BQP_{CTC} contains PSPACE, since we can simulate the classical time travel circuit for PSPACE using a quantum circuit. On the other hand, the best *upper* bound I know of on BQP_{CTC} is a class called SQG (Short Quantum Games), which was defined by Gutoski and Watrous [44] and which generalizes QIP (the class of problems having quantum interactive proof protocols). Note that QIP contains but is not known to equal PSPACE. Proving that $\text{BQP}_{\text{CTC}} \subseteq \text{SQG}$, and hopefully improving on that result to pin down the power of BQP_{CTC} exactly, are left as exercises for the reader.

8.1 The Algorithms of Bacon and Brun

My goal above was to explore the computational power of time travel in a clear, precise, complexity-theoretic way. However, there are several other perspectives on time travel computing; two were developed by Bacon [10] and Brun [25].

I assumed before that we have access to only one CTC, but can send a polynomial number of bits (or qubits) around that curve. Bacon considers a different model, in which we might be able to send only *one* bit around a CTC, but can use a polynomial number of CTC's. It is difficult to say which model is the more reasonable!

Like me, Bacon assumes Deutsch's causal consistency requirement. Bacon's main observation is that, by using a CTC, we could implement a 2-qubit gate similar to the nonlinear gates of Abrams and Lloyd [6], and could then use this gate to solve NP-complete problems in polynomial time. Even though Bacon's gate construction is quantum, the idea can be described just as well using classical probabilities. Here is how it works: we start with a chronology-respecting bit x , as well as a CTC bit y . Then a 2-bit gate G maps x to $x \oplus y$ (where $\oplus$ denotes exclusive OR) and y to x . Let $p = \Pr[x = 1]$ and $q = \Pr[y = 1]$; then causal consistency around the CTC implies that $p = q$. So after we apply G , the chronology-respecting bit will be 1 with probability

$$p' = \Pr[x \oplus y = 1] = p(1 - q) + q(1 - p) = 2p(1 - p).$$

Notice that if $p = 0$ then $p' = 0$, while if p is nonzero but sufficiently small then $p' \approx 2p$. It follows that, by applying the gate G a polynomial number of times, we can distinguish a bit that is 0 with certainty from a bit that is 1 with positive but exponentially small probability. Clearly such an ability would let us solve NP-complete problems efficiently.¹⁸ To me, however, the most interesting aspect of Bacon's paper is that he shows how standard quantum error-correction methods could be applied to a quantum computer with CTC's, in order to make his algorithm for solving NP-complete problems resilient against the same sort of noise that plagues ordinary quantum computers. This seems to be much easier with CTC quantum computers than with nonlinear quantum computers as studied by Abrams and Lloyd. The reason is that CTC's create nonlinearity *automatically*; one

¹⁸Indeed, in the quantum case one could also solve #P-complete problems, using the same trick as with Abrams and Lloyd's nonlinear gates.

does not need to build it in using unreliable gates.

Brun [25] does not specify a precise model for time travel computing, but from his examples, I gather that it involves a program computing a partial result and then sending it back in time to the beginning of the program, whereupon another partial result is computed, and so on. By appealing to the need for a “self-consistent outcome,” Brun argues that NP-complete as well as PSPACE-complete problems are solvable in polynomial time using this approach. As pointed out by Bacon [10], one difficulty is that it is possible to write programs for which there *is* no self-consistent outcome, or rather, no deterministic one. I also could not verify Brun’s claim to solve a PSPACE-complete problem (namely Quantified Boolean Formulas) in polynomial time. Indeed, since deciding whether a polynomial-time program has a deterministic self-consistent outcome is in NP, it would seem that PSPACE-complete problems *cannot* be solvable in this model unless $\text{NP} = \text{PSPACE}$.

Throughout this section, I have avoided obvious questions about the physicality of closed time-like curves. It is not hard to see that CTC’s would have many of the same physical effects as nonlinearities in quantum mechanics: they would allow superluminal signalling, the violation of Heisenberg’s uncertainty principle, and so on. As pointed out to me by Daniel Gottesman, there are also fundamental ambiguities in explaining what happens if half of an entangled quantum state is sent around a CTC, and the other half remains in a chronology-respecting region of spacetime.

9 “Anthropic Computing”

There is at least one foolproof way to solve 3SAT in polynomial time: given a formula φ , guess a random assignment x , then kill yourself if x does not satisfy φ . Conditioned on looking at anything at all, you will be looking at a satisfying assignment! Some would argue that this algorithm works even better if we assume the many-worlds interpretation of quantum mechanics. For according to that interpretation, with probability 1, there *really is* a universe in which you guess a satisfying assignment and therefore remain alive. Admittedly, if φ is unsatisfiable, you might be out of luck. But this is a technicality: to fix it, simply guess a random assignment with probability $1 - 2^{-2n}$, and do nothing with probability 2^{-2n} . If, after the algorithm is finished, you find that you have not done anything, then it is overwhelmingly likely that φ is unsatisfiable, since otherwise you would have found yourself in one of the universes where you guessed a satisfying assignment.

I propose the term “anthropic computing” for any model of computation in which the probability of one’s own existence might depend on a computer’s output. The name comes from the *anthropic principle* in cosmology, which states that certain things are the way they are because if they were different, then we would not be here to ask the question. Just as the anthropic principle raises difficult questions about the nature of scientific explanations, so anthropic computing raises similar questions about the nature of computation. For example, in formulating a model of computation, should we treat the user who picks an input x as an unanalyzed, godlike entity, or as part of the computational process itself?¹⁹

The surprising part is that anthropic computing leads not only to philosophical questions, but to nontrivial technical questions as well. For example, while it is obvious that we could solve NP-complete problems in polynomial time using anthropic postselection, could we do even more? Classically, it turns out that we could solve exactly the problems in a class called BPP_{path} , which was defined by Han, Hemaspaandra, and Thierauf [45] and which sits somewhere between MA and BPP^{NP} . The exact power of BPP_{path} relative to more standard classes is still unknown. Also, in a recent paper [5] I defined a quantum analogue of BPP_{path} called PostBQP . This class consists

¹⁹The same question is also asked in the much more prosaic setting of *average-case complexity* [54].

of all problems solvable in quantum polynomial time, given the ability to measure a qubit with a nonzero probability of being $|1\rangle$ and *postselect* on the measurement outcome being $|1\rangle$. I then showed that $\text{PostBQP} = \text{PP}$, and used this fact to give a simple, quantum computing based proof of Beigel, Reingold, and Spielman’s celebrated result [15] that PP is closed under intersection.

10 Discussion

Many of the deepest principles in physics are impossibility statements: for example, no superluminal signalling and no perpetual motion machines. What intrigues me is that there is a two-way relationship between these principles and proposed counterexamples to them. On the one hand, every time a proposed counterexample fails, it increases our confidence that the principles are really correct, especially if the counterexamples *almost* work but not quite. (Think of Maxwell’s Demon, or of the subtle distinction between quantum nonlocality and superluminal communication.) On the other hand, as we become more confident of the principles, we also become more willing to use them to constrain the search for new physical theories. Sometimes this can lead to breakthroughs: for example, Bekenstein [16] discovered black hole entropy just by taking seriously the impossibility of entropy decrease.

So, should the “NP Hardness Assumption”—loosely speaking, that NP-complete problems are intractable in the physical world—eventually be seen as a principle of *physics*? In my view, the answer ought to depend on (1) whether there is good evidence for the assumption, and (2) whether accepting it places interesting constraints on new physical theories. Regarding (1), we have seen that special relativity and quantum mechanics tend to support the assumption: there are plausible-sounding arguments for why these theories should let us solve NP-complete problems efficiently, and yet they do not, at least in the black box model. For the arguments turn out to founder on nontrivial facts about physics: the energy needed to accelerate to relativistic speed in one case, and the linearity of quantum mechanics in the other. As for (2), if we accept the NP Hardness Assumption, then presumably we should also accept the following:

- There are no nonlinear corrections to the Schrödinger equation, not even (for example) at a black hole singularity.²⁰
- There are no closed timelike curves.
- Real numbers cannot be stored with unlimited precision (so in particular, there should be a finite upper bound on the entropy of a bounded physical system).
- No version of the anthropic principle that allows arbitrary conditioning on the fact of one’s own existence can be valid.

These are not Earth-shaking implications, but neither are they entirely obvious.

Let me end this article by mentioning three objections that could be raised against the NP Hardness Assumption. The first is that the assumption is ill-defined: what, after all, does it *mean* to solve NP-complete problems efficiently? To me this seems like the weakest objection, since it is difficult to think of a claim about physical reality that is *more* operational. Most physical assertions come loaded with enough presuppositions to keep philosophers busy for decades, but the NP Hardness Assumption does not even presuppose the existence of matter or space. Instead

²⁰Horowitz and Maldacena [48] recently proposed such a modification as a way to resolve the black hole information loss paradox. See also a comment by Gottesman and Preskill [42].

it refers directly to information: an input that you, the experimenter, freely choose at time t_0 ,²¹ and an output that you receive at a later time t_1 . The only additional concepts needed are those of probability (in case of randomized algorithms), and of waiting for a given proper time $t_1 - t_0$. Naturally, it helps if there exists a being at t_1 who we can identify as the time-evolved version of the “you” who chose the input at t_0 !

But what about the oft-repeated claim that asymptotic statements have no relevance for physical reality? This claim has never impressed me. For me, the statement “*Max Clique requires exponential time*” is simply shorthand for a large class of statements involving reasonable instance sizes (say 10^8) but astronomical lengths of time (say 10^{80} seconds). If the complexity of the maximum clique problem turned out implausibly to be 1.000000001^n or n^{10000} , then so much the worse for the shorthand; the finite statements are what we actually cared about anyway. With this in mind, we can formulate the NP Hardness Assumption concretely as follows: “Given an undirected graph G with 10^8 vertices, there is no physical procedure by which you can decide in general whether G has a clique of size 10^7 , with probability at least $2/3$ and after at most 10^{80} seconds as experienced by you.”

The second objection is that, even if the NP Hardness Assumption *can* be formulated precisely, it is unlike any other physical principle we know. How could a statement that refers not to the flat-out impossibility of a task, but just to its probably taking a long time, reflect something fundamental about physics? On further reflection, though, the Second Law of Thermodynamics has the same character. The usual n particles in a box *will* eventually cluster on one side; it will just take expected time exponential in n . Admittedly there is one difference: while the Second Law rests on an elementary fact about statistics, the NP Hardness Assumption rests on some of the deepest conjectures ever made, in the sense that it could be falsified by a purely mathematical discovery such as $P = NP$. So as a heuristic, it might be helpful to split the Assumption into a ‘mathematical’ component ($P = NP$, $NP \not\subseteq BQP$, and so on), and a ‘physical’ component (there is no physical mechanism that achieves an exponential speedup for black-box search).

The third objection is the most interesting one: why NP? Why not PSPACE or #P or Graph Isomorphism? More to the point, why not assume *factoring* is physically intractable, thereby ruling out even garden-variety quantum computers? My answer is contained in the intuitive map shown in Figure 3. I will argue that, while a fast algorithm for graph isomorphism would be a mathematical breakthrough, a fast algorithm for inverting one-way functions, breaking pseudorandom generators, or related problems²² would be an almost *metaphysical* breakthrough.

Even many computer scientists do not seem to appreciate how different the world would be if we could solve NP-complete problems efficiently. I have heard it said, with a straight face, that a proof of $P = NP$ would be important because it would let airlines schedule their flights better, or shipping companies pack more boxes in their trucks! One person who did understand was Gödel. In his celebrated 1956 letter to von Neumann (see [68]), in which he first raised the P versus NP question, Gödel says that a linear or quadratic-time procedure for what we now call NP-complete problems would have “consequences of the greatest magnitude.” For such a procedure “would clearly indicate that, despite the unsolvability of the Entscheidungsproblem, the mental effort of the mathematician in the case of yes-or-no questions could be completely replaced by machines.”

But it would indicate even more. If such a procedure existed, then we could quickly find the

²¹Of course, your “free will” to choose an input is no different in philosophical terms from an experimenter’s “free will” to choose the initial conditions in Newtonian mechanics. In both cases, we have a claim about an infinity of possible situations, most of which will never occur.

²²Strictly speaking, these problems are “almost” NP-complete; it is an open problem whether they are complete under sufficiently strong reductions. Both problems are closely related to approximating the Kolmogorov complexity of a string or the circuit complexity of a Boolean function [8, 51].



Figure 3: My intuitive map of the complexity universe, showing a much larger gap between “structured” and “unstructured” problems than within either category. Needless to say, this map does not correspond to anything rigorous.

smallest Boolean circuits that output (say) a table of historical stock market data, or the human genome, or the complete works of Shakespeare. It seems entirely conceivable that, by analyzing these circuits, we could make an easy fortune on Wall Street, or retrace evolution, or even generate Shakespeare’s 38th play. For broadly speaking, that which we can compress we can understand, and that which we can understand we can predict. Indeed, in a recent book [12], Eric Baum argues that much of what we call ‘insight’ or ‘intelligence’ simply means finding succinct representations for our sense data. On his view, the human mind is largely a bundle of hacks and heuristics for this succinct-representation problem, cobbled together over a billion years of evolution. So if we could solve the general case—if knowing something was tantamount to knowing the shortest efficient description of it—then we would be almost like gods. The NP Hardness Assumption is the belief that such power will be forever beyond our reach.

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